Sriram Akella

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I am a final-year Ph.D. student at the Tata Institute of Fundamental Research, Mumbai, advised by Prof. Abhijit Gadde. I study multipartite entanglement and what it reveals about topological phases of matter, quantum error-correcting codes, and holographic spacetimes. More recently, I have begun exploring what nonlocal magic diagnoses in holographic and quantum many-body systems.

Education

- **Tata Institute of Fundamental Research**, Mumbai, India. 2021–present
  Integrated M.Sc.–Ph.D. in Physics. (**Advisor:** Prof. Abhijit Gadde)
  Expected Ph.D. completion: July 2027.
- **Chennai Mathematical Institute**, Chennai, India. 2018–2021
  B.Sc. (Hons.) in Mathematics and Physics.

Research Interests

Multipartite entanglement · topological order and TQFT · stabilizer codes and quantum error correction · holography and tensor networks · replica methods · quantum magic.

Preprints and Publications

Authors of papers 1–4 are listed alphabetically, following the hep-th convention. Also on INSPIRE.

1. **Akella, S.**, Iizuka, N. and Miyata, A., 2026. *Genuine Multi-Entropy in the Toric Code.* arXiv:2607.06050 [hep-th].
2. **Akella, S.** and Iizuka, N., 2026. *Structural Obstruction to Replica Symmetry Breaking for Multi-Entropy in Random Tensor Networks.* arXiv:2604.13261 [hep-th].
3. **Akella, S.**, Gadde, A. and Pandey, J., 2026. *Multi-invariants in stabilizer states.* arXiv:2601.16258 [quant-ph].
4. **Akella, S.**, 2025. *Tripartite entanglement in the HaPPY code is not holographic.* arXiv:2510.08520 [hep-th].
5. **Akella, S.**, Thapliyal, K., Mani, H.S. and Pathak, A., 2022. *Dynamics of single-mode nonclassicalities and quantum correlations in the Jaynes–Cummings model.* Journal of the Optical Society of America B, 39(7), pp. 1829–1838. arXiv:2203.04119 [quant-ph].

In Preparation

- **Akella, S.** and Gadde, A. *Rényi multi-entropy of graph states and the $n \to 1$ limit.*
- **Akella, S.** and Gadde, A. *Signals of genuine multiparty entanglement and topology.*
- **Akella, S.**, Iizuka, N. and Miyata, A. *Genuine Multi-Entropy in the Double Semion Theory.*

Talks and Lectures

- **Genuine multiparty entanglement in stabilizer states and topology.** 30 Sep 2026
  Taiwan String Workshop 2026
  National Tsing Hua University, Hsinchu.
- **Genuine Multi-Entropy in the Toric Code** 30 Jul 2026
  Yukawa Institute of Theoretical Physics, Kyoto University.
- **ST^4 Lectures on Quantum Information in Holography** 22 Jun - 4 Jul 2026
  Lecture series at the ST^4 workshop for advanced graduates and postdoctoral researchers in India.
  Chennai Mathematical Institute, Chennai, India.
- **Tripartite entanglement in the HaPPY code is not holographic** 4 Dec 2025
  Taiwan String Workshop 2025
  National Tsing Hua University, Hsinchu.
- **Virasoro TQFT as a description of pure 3d gravity** 21 Aug 2025
  Weekly seminar
  Tata Institute of Fundamental Research, Mumbai.

Research Visits

- **Yukawa Institute for Theoretical Physics** 10 Jul – 1 Aug 2026
  Kyoto University, Kyoto, Japan.

Schools, Workshops, Conferences

- **W.E. Heraeus Summer School "Saalburg" 2025.** Bayrischzell, Germany.
- **Quantum Information, Quantum Field Theory and Gravity 2024.** International Centre for Theoretical Sciences (ICTS-TIFR), Bengaluru, India.
- **Indian Strings Meeting 2023.** Indian Institute of Technology Bombay, Mumbai, India.
- **Strings 2023** and **It from Qubit 2023** (online). Perimeter Institute for Theoretical Physics, Waterloo, Canada.

Teaching

- **Quantum Field Theory 1** Aug–Dec 2024
  Teaching Assistant, Tata Institute of Fundamental Research.
- **Mathematical Methods** Aug–Dec 2022
  Teaching Assistant, Tata Institute of Fundamental Research.
- **Classical Mechanics** Aug–Dec 2020
  Teaching Assistant, Chennai Mathematical Institute.

Computational Skills

Python, Julia, Mathematica, LaTeX, Typst.

References

Prof. Abhijit Gadde (Ph.D. Advisor)
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